

#3 ICI vulnerability dark thyroid cancer

*"HLA loss, neoantigen architecture, and immune ecotypes define ICI vulnerability in molecularly dark thyroid cancer" – Track A 디자인 phase 가 **chmod 444 frozen** 상태이고, Track B 실행은 (1) Paper 1 bioRxiv + (2) Paper 2B Pillar II/III closure + (3) literal "Paper 3 Track B 시작" 명령 모두 충족 시 unlock.*

FROZEN

Track A: chmod 444 (2026-05-04)

Track B: BLOCKED

Bundle: 8 deliverables · 1,236 lines

tar.gz: 51 KB · sha256 290349f6...8936aa

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I. 한 줄 결론

Paper 3 = FROZEN. Track A (디자인 phase) chmod 444 frozen – 8 deliverable + tar.gz 51 KB + sha256 290349f6...8936aa . **Design bundle 위치** = project/reports/paper3_ici/ PAPER3_ICI_TRACK_A_BUNDLE.md (chmod 444). Track B 실행은 (1) **Paper 1** bioRxiv + (2) **Paper 2B** Pillar II/III closure + (3) literal "Paper 3 Track B 시작" 명령 모두 충족 시 unlock. "고고" / "다 해줘" / "faster" 명령은 Track B 를 *unlock* 하지 않는다.

TRACK A STATUS

FROZEN

chmod 444 ·
2026-05-04 · marathon-
binding

BUNDLE SCALE

1,236

8
deliverables
· single-file
md ·
71,684
bytes

TAR ARCHIVE

**51
KB**

sha256
290349f6...
8936aa

TRACK B

BLOCKED

3-condition unlock; literal
phrase only

PROBLEM

BRAF/RAS-
negative dark
matter PTC +
ATC/PDTC 의
면역치료 후보
정의 자체가
unsolved –
thyroid 의 ICI
raw cohort 가
전세계 부재.

IDEA

HLA loss /
neoantigen
architecture /
immune
ecotype 의 분
자 marker 만으
로 ICI
vulnerability
profile 을
stratify;
dataset 부재 상
황에서 책임감
있는 framing.

DEFENSE

"ICI response
predictor" 단
어 사용 시 즉
시 paper kill
(K1). 대신 "ICI
vulnerability" /
"ICI-
readiness" /
"immunogenomic
prioritization"
만 허용. K1-
K6 + G1-G6
binding.

DECISION

Track A
chmod 444
frozen, Track
B BLOCKED
– Paper 1
bioRxiv +
Paper 2B
closure + 명시
적 "Paper 3
Track B 시작"
명령 모두 충족
시 chmod 644
로 변경.

2. 비전공자 배경

전체 PTC 의 ~95% 는 surgical resection + RAI 치료로 cure 되지만 ~4% 는 high-risk recurrent / metastatic 으로 진행. 이 4% 안에 (a) **BRAF V600E + TERT** driver-positive aggressive type, (b) **BRAF/RAS-negative dark matter** driver-unknown type 이 같이 있다. Driver-positive 는 BRAF inhibitor (dabrafenib + trametinib) 등 targeted therapy 가 후보. **Driver-negative dark matter** 환자는 targeted therapy 후보 없음 – 면역치료 (PD-1 / PD-L1 inhibitor) 가 마지막 옵션. 그러나 thyroid 는 (a) tumor mutational burden 평균 낮음, (b) MHC-I / MHC-II 발현 cohort heterogeneity, (c) ICI clinical trial 매우 적음 – ICI 후보 집단 정의 자체가 unsolved.

"Response predictor" 는 ICI 받은 환자의 outcome 을 예측하는 model – 이걸 만들려면 **ICI-treated thyroid cancer cohort raw RNA-seq + treatment outcome** 데이터가 필요. 전세계 부재. 따라서 어떤 signature 가 thyroid 의 ICI response 를 예측한다고 못 한다. 대신 "**vulnerability profile**" = HLA loss / neoantigen burden / immune ecotype 의 분자 marker 만으로 ICI 후보 prioritize. 이걸 published immune-oncology 의 framework 안에서 (TIDE Jiang 2018 / IMPRES Auslander 2018 / TIS Ayers 2017 / etc.) thyroid-specific 적용.

2.I 용어 정리

TERM	풀어쓴 설명	PAPER 3 에서의 역할
Track A	디자인 phase – 8 deliverable (hypothesis / cohort / methods / analysis / DIAL / pre-reg / gates+kills / boundary).	chmod 444 frozen 2026-05-04.
Track B	12-week 실행 phase – Wk 1 verify+ETL → Wk 10 integrated score → Wk 11–12 manuscript draft.	BLOCKED until 3-condition unlock.
DIAL audit	Direction-Invariance Algorithm – pan-cancer ICI signature 가 thyroid 로 transfer 시 sign 유지 여부 검증.	4-verdict matrix: PASS / FLIP / COLLAPSE / AMBIGUOUS.
K1 kill switch	"ICI response predictor" claim 사용 시 즉시 paper kill – raw thyroid ICI cohort 부재.	가장 중요한 boundary; reviewer Q1 사전 답변.
cookHLA	Cook 2021 Nat Commun cross-disease HLA imputation framework.	Reserved for Paper 4. Paper 3에서는 OptiType / Polysolver / arcasHLA / xHLA 만.
v52_lodo_finding	v5.1 THCA DIAL=0.494 flip 이 leak artifact, LODO ComBat 하면 모두 0.000.	K2 kill – 그 number 다시 인용 시 즉시 kill.

3. 전체 논리 흐름

Paper 3 의 이야기는 "ICI vulnerability 라는 strong concept 을 Track A design freeze 로 보존하되, marathon 동안 active work 차단; Track B 시작 조건 모두 binding". 이 paper 자체가 [Paper 7](#) Component B (chmod 444 freeze) archetype.

- 1 **BRAF/RAS-negative dark matter PTC + ATC/PDTC 를 ICI vulnerability 후보 집단으로 정의.** Driver-negative 환자는 targeted therapy 후보 없음 – 면역치료가 마지막 옵션.
- 2 **K1 kill switch 정의.** "ICI response predictor" 단어 사용 금지 – raw thyroid ICI cohort 부재. 대신 "ICI vulnerability" / "ICI-readiness" / "immunogenomic prioritization".
- 3 **Track A 8 deliverable 작성 + chmod 444 freeze.** Hypothesis spec / Cohort access plan / Methods / Analysis / DIAL Audit Plan / Pre-reg / G1-G6 + K1-K6 / Boundary statement.
- 4 **tar.gz + sha256 verification.** 51 KB tar.gz · sha256 290349f6... 8936aa – marathon 동안 design 보호.
- 5 **Editable todo (chmod 644) 만 외부 –** Track B kickoff checklist + controlled-access checklist + dataset accession verification.
- 6 **Track B unlock 조건 3개 모두 binding.** Paper 1 bioRxiv + Paper 2B Pillar II/III closure + literal "Paper 3 Track B 시작" 명령. literal phrase only.

3.I 분석 사고 스토리

Q1 - 왜 ICI VULNERABILITY 인가?

BRAF/RAS-negative dark matter
는 targeted therapy 후보 없음. ICI
가 마지막 옵션이지만 thyroid ICI
cohort 부재 – vulnerability
profile 로 후보 prioritize.

**Q2 - 왜 "PREDICTOR" 가 아닌
"VULNERABILITY" 인가?**

Predictor 는 outcome 예측 → raw
cohort 필요. Vulnerability 는 분자
marker – dataset 부재 상황에서
책임감 있는 framing.

Q3 - 왜 CHMOD 444 FROZEN 인가?

Track A 디자인이 manuscript
backbone – marathon 중 마음대
로 수정 시 (a) reviewer Q1 사전답
변 흔들림, (b) DIAL audit plan
post-hoc selection bias, (c) Paper
1 마라톤 displacement.

Q4 - 왜 COOKHLA 안 쓰나?

cookHLA 는 Paper 4 reserved
(Korean GD HLA backlog). Paper
3 는 OptiType / Polysolver /
arcasHLA / xHLA 4-tool
consensus.

Q5 - 왜 K1 가장 강한 KILL 인가?

"ICI response predictor" 사용 시
reviewer 가 첫 page 에서 reject –
raw cohort 부재 inconsistency 가
즉시 노출.

**FINAL - TRACK A FROZEN, TRACK
B BLOCKED.**

Paper 1 ship + Paper 2B closure +
literal "Paper 3 Track B 시작" 명령
모두 충족 시에만 chmod 644
unlock.

4. FROZEN bundle 위치

Paper 3 design bundle 은 read-only 상태에서 Track B unlock 까지 유지. 임의의 chmod 변경 또는 design bundle 수정 금지. literal "Paper 3 Track B 시작" + 3 unlock 조건 모두 필요.

ITEM	PATH / VALUE
Single-file consolidated bundle (chmod 444)	project/reports/paper3_ici/ PAPER3_ICI_TRACK_A_BUNDLE.md · 1,236 lines · 71,684 bytes
Tar archive (chmod 444)	project/reports/paper3_ici/ paper3_ici_track_a.tar.gz · 51,023 bytes · sha256 290349f6...8936aa
Freeze notice	project/reports/ PAPER3_ICI_TRACK_A_FREEZE_2026_05_04.md
Decision brief	project/reports/ PAPER3_ICI_DECISION_BRIEF_2026_05_04.md (A/B/C tree)
8 individual designs (chmod 444)	project/reports/paper3_ici/paper3_ici_*.md
Editable todo (chmod 644)	project/reports/paper3_ici/todo/ · 3 checklists

4.1 Track B unlock 조건 (3-of-3 binding)

#	CONDITION	CURRENT	TARGET DATE
1	Paper 1 bioRxiv 제출 완료	 6/13 target	2026-06-13
2	Paper 2B Pillar II/III closure (Yu 미팅 input 후)	 pending	~2026-07-15
3	literal "Paper 3 Track B 시작" 명령 (사용자 직접)	 미발화	위 두 충족 후

"고고" / "다 해줘" / "faster" 명령은 Track B 를 unlock 하지 않는다. literal phrase 만 인정.

4.2 K-V 빠른 요약

KEY	VALUE
Status	Track A FROZEN 2026-05-04 (chmod 444 bundle + tar.gz). Track B BLOCKED.
Allowed claim	"ICI vulnerability" / "ICI-readiness" / "immunogenomic prioritization"
Forbidden claim	"ICI response predictor" – kill switch K1 (raw cohort 부재)
Track B unlock	(1) Paper 1 bioRxiv 제출 + (2) Paper 2B Pillar II/III closure + (3) literal "Paper 3 Track B 시작" command
Memory	v19_paper3_ici_track_a · v17_landa2016_cite_save · v17_K2_vs_bundang_distinction

5. Track A — 8 deliverables (chmod 444 design bundle)

#	DELIVERABLE	핵심
1	Dataset Registry	bulk ≤ 9 / scRNA ≤ 6 / spatial ≤ 4 / pan-cancer ICI ≤ 7 . 모두 <code>to_verify</code> 상태 (Track B Wk 1 portal lookup).
2	Signature Registry	IFN γ / TIS / Cytolytic / MHC-I/II / TLS / CXCL13 / Myeloid / Treg / TIDE-like / IMPRES + tumor-intrinsic (BRS / ERK / TDS / lineage) + sc-derived placeholder. Module E 가중치 deferred.
3	scRNA Atlas Plan	scVI / scANVI / Harmony 통합. Level-1/2 taxonomy. sc-derived signature lock for bulk projection. PDTC/ATC < 5K caveat.
4	HLA & Neoantigen Feasibility	OptiType / Polysolver / arcasHLA / xHLA + LOHHLA + NetMHCpan/NetMHCIIpan. cookHLA reserved for Paper 4 . dbGaP unauthorized 시 degraded scope contingency.
5	DIAL Audit Plan	sign-consistency $\times$ effect-size $\times$ tissue-transfer $\rightarrow$ PASS / FLIP / COLLAPSE / AMBIGUOUS verdict per signature. Integrated thyroid-readiness score = PASS only.
6	Figure Plan	F1 cohort/dark matter / F2 ecotype NMF / F3 sc atlas / F4 HLA + neoantigen / F5 DIAL / F6 integrated score; 14 supp.
7	Go/No-Go Verdict	Conditional GO. Hard gates G1–G6 + kill switches K1–K6.
8	12-Week Execution Plan	Wk 1 verify+ETL $\rightarrow$ Wk 10 integrated score $\rightarrow$ Wk 11–12 manuscript draft.

5.1 8 deliverables — scope 확장 (chmod 444 frozen)

#	DELIVERABLE	SCOPE DETAIL	TRACK B 시간
1	Dataset Registry	bulk ≤ 9 (TCGA-THCA / Lee 2024 / GSE76039 ATC / GSE33630 등) · scRNA ≤ 6 (Pu 2021 / Hu 2024 / Han 2023) · spatial ≤ 4 (GSE250521 / Lee STAR 등) · pan-cancer ICI ≤ 7 (Liu 2019 / Mariathasan 2018 / Riaz 2017 / Hugo 2016 / Van Allen 2015)	Wk 1 verify
2	Signature Registry	11 immune-cohort signatures (IFN γ Ayers 2017 / TIS / Cytolytic / MHC-I/II / TLS Cabrita / CXCL13 / Myeloid / Treg / TIDE Jiang 2018 / IMPRES Auslander 2018) + 4 tumor-intrinsic (BRS / ERK / TDS / lineage) + sc-derived (TBD via scRNA atlas)	Wk 2 lock
3	scRNA Atlas Plan	scVI / scANVI / Harmony 통합 · Level-1 (epithelial / immune / stromal / endothelial) + Level-2 (T CD8 / T CD4 / Treg / B / Plasma / Myeloid / DC / Macrophage / Fibroblast) · sc-derived signature lock for bulk projection	Wk 3-4
4	HLA & Neoantigen Feasibility	4 HLA imputation (OptiType / Polysolver / arcasHLA / xHLA) consensus + LOHHLA + NetMHCpan / NetMHCIIpan epitope prediction · cookHLA reserved for Paper 4	Wk 5-6
5	DIAL Audit Plan	sign-consistency $\times$ effect-size $\times$ tissue-transfer 3차 원 verdict matrix $\rightarrow$ PASS / FLIP / COLLAPSE / AMBIGUOUS per signature · pan-cancer ICI cohort 에서 thyroid 로 transfer 시 sign 유지 여부	Wk 7
6	Figure Plan	F1 cohort / dark matter / F2 ecotype NMF / F3 sc atlas integration / F4 HLA loss + neoantigen burden / F5 DIAL audit verdict / F6 integrated readiness score · 14 supp	Wk 8-10
7	Go/No-Go Verdict	Conditional GO. Hard gates G1–G6 + kill switches K1–K6	Wk 11
8	12-Week Execution Plan	Wk 1 verify+ETL $\rightarrow$ Wk 10 integrated score $\rightarrow$ Wk 11–12 manuscript draft	—

5.2 Editable todo (chmod 644, freeze 외부)

CHECKLIST	PURPOSE
Track B kickoff checklist	G1-G6 gate 확인 + design-bundle re-read + environment readiness (Track B kickoff trigger)
Controlled access checklist	dbGaP / EGA application · ≤30 min/session marathon-displacement-safe · background-only during marathon
Dataset accession verification	Portal-lookup verification of all <code>to_verify</code> accessions (Track B Wk 1 default deferred)

6. Gates G_I–G₆ / Kills K_I–K₆ (binding)

6.1 Hard gates G_I–G₆ (요약)

GATE	THRESHOLD	IF FAIL
G1 Cohort verify	≥80% of registry datasets accessible	scope 좁힘 (degraded)
G2 Signature DIAL PASS rate	≥6/15 signatures PASS in thyroid	integrated score 정의 못함
G3 scRNA integration QC	silhouette > 0.4 across batches	scRNA atlas drop, bulk only
G4 HLA imputation pass rate	≥80% samples 4-digit HLA-A/B/C	HLA module 약화
G5 Neoantigen burden distribution	≥10 epitope/sample median	neoantigen panel drop
G6 Integrated readiness score reproducibility	cross-cohort $r \geq 0.6$	integrated score 못 publish

6.2 Hard gates G_I–G₆ — Track B 시작 조건 detail

GATE	CONDITION	QUANTITATIVE CRITERION	VERIFIER
G1	Paper 1 bioRxiv submitted	DOI assigned, manuscript public	Yu advisor
G2	Paper 2B Pillar I/II/III closure	Pillar II + III scope defined + analysis complete	Yu advisor
G3	ICI cohort access secured	≥1 ICI-treated thyroid cohort with paired RNA-seq + clinical outcome	cohort source IRB approval
G4	Yu advisor green light	Explicit "Paper 3 Track B 시작" 명령	Yu lab notebook record
G5	Funding alignment	Compute + cohort access budget secured	grant statement
G6	Pre-registration filed	OSF / GitHub release of paper3_prereg_template_v1.md filled	OSF DOI
모든 G1–G6 충족 시 Track B 시작 권한 발생. 한 개라도 미충족 시 Track B 차단 binding.			

6.3 Kill switches K_I–K₆ (요약)

KILL	TRIGGER	이유
K1	"ICI response predictor" claim	raw thyroid ICI cohort 부재 – predictor 라고 못 함
K2	scRNA < 5K cell PDTC/ ATC	sc-derived signature unstable
K3	Signature DIAL FAIL rate > 50%	integrated score 의 base 약함
K4	HLA imputation < 80%	HLA module 의 cohort representation 부족
K5	TIDE-like sign-flip in thyroid	pan-cancer signature 가 thyroid 에서 opposite – 신뢰성 없음
K6	Cohort size < 50 dark matter	statistical power 부족

6.4 Kill switches K_I–K₆ — Track B 즉시 kill 조건 (detail)

KILL	CONDITION	TRIGGER	ACTION
K1	"ICI response predictor" framing 등장	any draft / submission text	즉시 paper kill (vulnerability ≠ response)
K2	v15 v5.1 DIAL = 0.494 number cite (post-v52 lock)	any text / supplement	즉시 kill (leak number)
K3	Paper 1/2B scope contamination	image-DM1 / HT- overlap / DM1 4-step in Paper 3	즉시 kill (cross- paper boundary violation)
K4	DIAL audit verdict = COLLAPSE (across all classifiers)	analysis result	즉시 kill (no signal)
K5	Pre-registered Spearman gate (≥ 0.30 or BORDERLINE 0.20) 미달	analysis result	즉시 kill (insufficient signal)
K6	Functional / wet-lab requirement 우선 변경 시	Yu advisor pivot	즉시 kill (scope drift)
K1–K6 중 어느 하나라도 trigger 시 Track B execution 즉시 중단 + design bundle 재 freeze.			

7. Track A 8 deliverables — file inventory + sha256

#	DELIVERABLE	FILE	CHMOD	LINES	STATUS
1	Hypothesis spec	paper3_hypothesis_v1.md	444	~120	✓ frozen 2026-05-04
2	Cohort access plan	paper3_cohort_access_v1.md	444	~150	✓ frozen
3	Methods spec (immune deconv + DIAL audit)	paper3_methods_spec_v1.md	444	~280	✓ frozen
4	Analysis spec	paper3_analysis_spec_v1.md	444	~210	✓ frozen
5	DIAL Audit Plan (4-verdict framework)	paper3_dial_audit_plan_v1.md	444	~180	✓ frozen (post-v52 lesson)
6	Pre-registration template	paper3_prereg_template_v1.md	444	~150	✓ frozen
7	Hard gates G1–G6 + Kill switches K1–K6	paper3_gates_kills_v1.md	444	~100	✓ frozen (binding)
8	Boundary statement + ICI vulnerability vs ICI response distinction	paper3_boundary_v1.md	444	~46	✓ frozen

Total bundle: 1,236 lines · sha256 = **290349f6 ... 8936aa** · packed as **paper3_track_a_bundle_2026_05_04.tar.gz** (~51 KB) · all chmod 444 read-only · marathon-binding.

7.1 Track A bundle inventory (file-by-file bytes / mode / role)

FILE	BYTES	MODE	ROLE
PAPER3_ICI_TRACK_A_BUNDLE.md	71,684	444	Single-file consolidated design (FROZEN marker prepended)
paper3_ici_dataset_registry.md	9,947	444	D1 – bulk / scRNA / spatial / pan-cancer ICI dataset registry
paper3_ici_signature_registry.md	9,887	444	D2 – signature registry + integrated score composition
paper3_ici_scRNA_reference_atlas_summary.md	7,958	444	D3 – scRNA atlas plan
paper3_ici_HLA_neoantigen_feasibility.md	8,717	444	D4 – HLA typing LOHHLA + NetMHCpan
paper3_ici_DIAL_audit_plan.md	8,383	444	D5 – DIAL audit plan
paper3_ici_figure_plan.md	8,570	444	D6 – 6 main + 14 supp figure plan
paper3_ici_go_no_go_verdict.md	5,794	444	D7 – Conditional GO + G1-G6 + K1-K6
paper3_ici_12week_execution_plan.md	11,602	444	D8 – 12-week Track B execution schedule
paper3_ici_track_a.tar.gz	51,023	444	Tar archive (sha256: 290349f6...8936a)
todo/paper3_ici_track_b_kickoff_checklist.md	4,895	644	Track B kickoff verification (editable)
todo/paper3_ici_controlled_access_checklist.md	4,708	644	dbGaP/EGA application (editable, ≤30 min session)
	6,234	644	

FILE	BYTES	MODE	ROLE
todo/ paper3_ici_dataset_accession_verification_checklist.md			Wk 1 portal look (editable)

8. Research narrative — Track A design freeze timeline

Paper 3 의 이야기는 "ICI vulnerability 라는 strong concept 을 Track A design freeze 로 보존하되, marathon 동안 active work 차단; Track B 시작 조건 모두 binding". 이 paper 자체가 [Paper 7](#) Component B (chmod 444 freeze) archetype.

DATE	PHASE	EVENT	DECISION / IMPACT
2026-03 ~ 04	v13 → v15 design phase	"ICI vulnerability dark thyroid cancer" 컨셉 정립 – DM1 dark matter 와 ICI response 의 연결 가설	Track A (design) vs Track B (execution) split decision
2026-04-30	Track A 8 deliverables 정의	(1) Hypothesis spec, (2) Cohort access plan, (3) Methods spec (immune deconvolution + DIAL audit), (4) Analysis spec, (5) DIAL Audit Plan (PASS/FLIP/COLLAPSE/AMBIGUOUS), (6) Pre-registration template, (7) Hard gates G1-G6 + Kill switches K1-K6, (8) Boundary statement	1236-line bundle + sha256 verification + tar.gz preservation
2026-05-04	chmod 444 freeze	Track A 8 deliverables 모두 chmod 444 (read-only) 적용; bundle sha256 = 290349f6...8936aa	Marathon 동안 design 보호; Paper 7 Component B archetype 으로 등재
2026-05-04	Editable todo (chmod 644)	Frozen 외부에 editable todo + Hard gates G1-G6 + Kill switches K1-K6 정의 – Track B 시작 조건 + 즉시 kill 조건 명시화	Track B unlock 의 binding gating; "ICI response predictor" 같은 K1 kill 영역 미리 차단
2026-05-04	This hub page	Paper 3 hub formal documentation; frozen status visible	marathon 동안 모든 active work 차단
2026-06-13 (target)	Condition: Paper 1 bioRxiv	Paper 1 bioRxiv submission	Track B unlock 의 1차 trigger
2026-Q3 ~ Q4 (estimate)	Track B 시작 조건 모두 충족 시	(a) Paper 1 bioRxiv, (b) Paper 2B closure, (c) ICI cohort access 확보, (d) Yu green light	Track B execution begin
		Track B 실행 후 ~6 months → manuscript drafting	Paper 3 active scope

DATE	PHASE	EVENT	DECISION / IMPACT
2027-Q1 ~ Q2 (estimate)	Track B → Paper 3 manuscript		

9. Target venue (post-Track B execution)

VENUE	IF (2024)	적합성	예상 ACCEPTANCE	이유
J Immunother Cancer (JITC)	~10.3	★★★★ ICI vulnerability 직속	medium-high	ICI biomarker 의 home journal; thyroid-specific 가능
Cancer Immunol Res	~7.5	★★★★ ICI mechanism	medium-high	Mechanism + biomarker; 우리 axis 의 perfect fit
Cell Rep Med	~13.0	★★★ translational	medium	If Bundang ICI cohort + functional layer 추가 시 reach
npj Precision Oncology	~7.0	★★★ precision oncology biomarker	medium-high	DIAL audit 가 precision biomarker framework 으로 fit
Mol Cancer	~28	★★ if mechanistic depth + cohort 통합 시	low without functional	Mechanism + biomarker + cohort 모두 통합 시 가능
Nat Cancer / Nat Med	~32-82	★ if pan-cancer + ICI clinical trial 통합 시	very low without trial	ICI clinical trial 의 prospective biomarker validation 동반 시만 가능

Realistic primary target (post-Track B): *JITC* (ICI biomarker home) 또는 *Cancer Immunol Res* (mechanism + biomarker fit). Reach: Cell Rep Med / npj Precision Oncology (translational layer 추가 시).

10. Track B 시작 + Paper 3 작성 위해
추가 필요한 데이터 (3 tiers)

TIER	WHAT	WHY	EFFORT	STATUS
Tier 1 – Critical (G1-G6 gates)	ICI-treated thyroid cancer cohort access (TCGA 외, e.g., MSKCC / MD Anderson published cohort 또는 Bundang prospective)	"ICI vulnerability" claim 의 evidence base; 현재 cohort 부재	6-12 months outreach + IRB	 post-marathon
Tier 1	DIAL Audit Plan 의 4-verdict framework (PASS / FLIP / COLLAPSE / AMBIGUOUS) implementation (v15 v5.1 number 와 분리)	v52 leak 학습 적용; Paper 6 의 lesson 위에 구축	2 months Python framework	 design ready
Tier 1	Immune deconvolution pipeline (CIBERSORT / TIDE / immunedeconv 통합)	ICI response 의 immune microenvironment 정량화	1 month integration	 design ready
Tier 2 – Strengthening	Spatial transcriptomics ICI-treated cohort (Visium 또는 GeoMx) – DM1 axis × immune spatial colocalization	Mechanism + spatial validation; Cell Rep Med reach	6 months prospective	 post-Tier 1
Tier 2	scRNA-seq ICI cohort – DM1 score × T cell exhaustion correlation	Single-cell resolution mechanism	3 months curation + analysis	 post-Tier 1
Tier 2				

TIER	WHAT	WHY	EFFORT	STATUS
	Functional in vitro ICI response (organoid + sacituzumab + anti-PD1 combo)	Wet-lab validation; high-impact venue 보강	6+ months wet-lab	 long-term
Tier 3 – Future	Prospective ICI clinical trial (Phase II investigator-initiated; DM1-stratified)	Nat Cancer / Nat Med tier; clinical translation	2+ years + IRB + funding	 grant ambition
Tier 3	Pan-cancer ICI vulnerability extension (BRCA / NSCLC / melanoma 에 framework 적용)	Generality 입증; thyroid-only 비판 차단	1+ year cross-cancer	 ambition

10.1 다음에 필요한 데이터 — Track B Wk 1 verify+ETL Tier 1

- thyroid PD-1/PD-L1 trial cohort raw RNA-seq + outcome**
(NCT04645966 / NCT03012620 / NCT02239900 등 – controlled access 신청).
- Pan-cancer ICI cohort dbGaP access** (Liu 2019 PD-1 atlas / Mariathasan 2018 IMvigor210 / Riaz 2017 anti-PD-1 / Hugo 2016 anti-PD-1).
- thyroid scRNA atlas integration** (Pu 2021 PMC8754658 + Hu 2024 + Han 2023).
- Spatial thyroid cancer atlas** (GSE250521 + 추후).

10.2 Tier 2 — DIAL audit + signature lock (Wk 2-7)

- 11 published immune signatures verified scores** on thyroid (TIDE / IMPRES / TIS / IFN γ -Ayers / etc.).

2. **scRNA-derived thyroid-specific signatures** (Module E weighting).
3. **Tumor-intrinsic signatures** (BRS / ERK / TDS / lineage) verified on thyroid.

10.3 Tier 3 — HLA + Neoantigen (Wk 5-6)

1. **4 HLA imputation consensus** (OptiType / Polysolver / arcasHLA / xHLA) on thyroid bulk.
2. **LOHHLA** for HLA loss-of-heterozygosity detection.
3. **NetMHCpan / NetMHCIIpan** epitope prediction from somatic mutations.

10.4 Track B 12-week execution plan (post-unlock)

WK	MODULE	ACTIVITY	HARD GATE
1	verify + ETL	Dataset Registry portal lookup, dbGaP/EGA application	G1: $\geq 80\%$ datasets accessible
2	Signature lock	11 immune + 4 tumor-intrinsic signatures verify	G2: $\geq 6/15$ signatures DIAL PASS
3-4	scRNA atlas	scVI / scANVI / Harmony integration · Level-1/2 taxonomy	G3: silhouette > 0.4
5-6	HLA + neoantigen	OptiType / Polysolver / arcasHLA / xHLA + LOHHLA + NetMHCpan	G4: $\geq 80\%$ HLA imputation
7	DIAL audit	sign-consistency $\times$ effect-size $\times$ tissue-transfer per signature	G5: $\geq 3/5$ ICI cohorts accessible
8	Module D pan-cancer ICI	Liu 2019 / Mariathasan 2018 / Riaz 2017 / Hugo 2016 transfer	(K5: TIDE-like sign-flip $\rightarrow$ narrow scope)
9-10	Integrated readiness score	Module E weighting calibration	G6: cross-cohort $r \geq 0.6$
11-12	Manuscript draft	F1-F6 main + 14 supp + Methods + cover letter	K1: never use "ICI predictor"

II. Dataset / Signature registry — frozen excerpts

II.I Dataset Registry — bulk + scRNA + spatial + pan-
cancer ICI

MODALITY	COHORT NAME	ACCESSION	N	STATUS	ROLE
bulk	TCGA-THCA	GDC: TCGA-THCA	500	verified	Discovery + dark matter
bulk	Lee 2024	GEO: GSE213647	632	verified	Korean replication
bulk	K2 PRJEB11591	ENA: PRJEB11591	260	verified	Korean PTC HLA
bulk	GSE286332	GEO: GSE286332	18	verified	PTC vs PTC+HT
bulk	GSE76039	GEO: GSE76039	~100	to_verify	ATC
bulk	GSE33630	GEO: GSE33630	~100	to_verify	PTC
bulk	GSE184362	GEO: GSE184362	~50	to_verify	Aggressive
scRNA	Pu 2021	PMC8754658	~30K	to_verify	scRNA atlas integration
scRNA	Hu 2024	(TBD)	~50K	to_verify	scRNA atlas
scRNA	Han 2023	(TBD)	~20K	to_verify	scRNA atlas
spatial	GSE250521	GEO: GSE250521	28 slides	verified	Spatial backbone
pan-cancer ICI	Liu 2019 PD-1 atlas	dbGaP	~120	to_verify	Signature transfer
pan-cancer ICI	Mariathasan 2018 IMvigor210	EGA	~300	to_verify	TIDE-like benchmark
pan-cancer ICI	Riaz 2017 anti-PD-1	dbGaP	~70	to_verify	Cohort transfer

II.2 Signature Registry — 11 immune + 4 tumor-intrinsic
+ sc-derived

TYPE	SIGNATURE	SOURCE	GENES	DIAL AUDIT VERDICT
immune	IFN- γ Ayers 2017	JCI 2017	~18 genes	to_audit
immune	TIS	Tumor inflammation signature	18 genes	to_audit
immune	Cytolytic	Rooney 2015	2 genes (GZMA + PRF1)	to_audit
immune	MHC-I module	HLA-A/B/C + B2M + TAP1/2	~6 genes	to_audit
immune	MHC-II module	HLA-DR/DP/DQ + CIITA + HLA-DM/DO	~10 genes	to_audit
immune	TLS Cabrita 2020	Nature 2020	12 genes	to_audit
immune	CXCL13	–	1 gene	to_audit
immune	Myeloid module	–	~15 genes	to_audit
immune	Treg module	FOXP3 + CTLA4 + IKZF2	~10 genes	to_audit
immune	TIDE-like Jiang 2018	Nat Med 2018	–	K5 risk: thyroid sign-flip 가능
immune	IMPRES Auslander 2018	Nat Med 2018	15 pairs	to_audit
tumor-intrinsic	BRS	BRAF-RAS score	–	to_audit
tumor-intrinsic	ERK	–	–	to_audit
tumor-intrinsic	TDS	Thyroid differentiation score	16 genes	to_audit
tumor-intrinsic	lineage	8-gene RAI panel (Paper 1)	8 genes	to_audit

12. 8 deliverable excerpts (chmod 444 read-only)

Frozen design bundle 의 8 deliverables 각각의 핵심 발췌. 원본은 chmod 444 read-only.

I. Dataset Registry · `paper3_ici_dataset_registry.md`

```
# Paper 3 ICI – Dataset Registry

**Working title:** HLA loss, neoantigen architecture, and
immune ecotypes define ICI vulnerability in molecularly dark
thyroid cancer
**Track:** A – design only. No download, no analysis. All
accessions marked `to_verify` until Track B kickoff.
**Author:** Seungho Cook
**Status date:** 2026-05-04

**Claim guard:** Datasets are catalogued for ICI
vulnerability / immunogenomic prioritization, never for "ICI
response prediction" in thyroid unless thyroid ICI-treated
raw RNA-seq is acquired.

---

## 0. Verification policy

For each accession:
- **`to_verify`** – accession string is from prior
literature / user input; existence, version, file types, and
access conditions must be checked at Track B kickoff via GEO/
SRA/EGA/dbGaP web UI (not by Claude during Track A).
- **`internal`** – already on local mirror or processed for
Paper 1/2; reuse read-only with paper-boundary discipline.
- **`controlled`** – known to be controlled-access (dbGaP /
EGA / institutional). Application timing must be added to the
12-week plan.
- **`tier_x`** – see priority tiers below.

Priority tiers:
- **Tier 1** – required for headline figure. Track B blocks
if unavailable.
- **Tier 2** – replication / cross-platform. Track B proceeds
if at least 50% of Tier 2 are usable.
- **Tier 3** – supplementary / sensitivity. Loss tolerable.
```

1. Bulk RNA-seq ± WES – discovery & dedifferentiation cohorts

Accession	Modality	n (claimed)	Subtype	Role	Tier	Status	Notes
TCGA-THCA	RNA-seq + WES + clinical	~500	PTC, FVPTC, TCV; ATC limited	Primary discovery: ecotype NMF, HLA LOH, neoantigen, dark-matter stratification	1	internal	Driver/fusion calls already curated in Paper 1 ETL. WES BAM available via GDC; LOHHLA needs raw BAM not just MAF.
GSE76039	RNA-seq	~37 (PDTC + ATC; Landa 2016)	PDTC, ATC	Dedifferentiation axis anchor	1	to_verify	Landa 2016 JCI (cite already saved in `v17_landat2016_cite_save`). Small n – pair with TCGA aggressive subset for power.
GSE33630	microarray (Affy U133 Plus 2)	~105	PTC, ATC, normal	Cross-platform replication	2	to_verify	Microarray, so signature scoring only – no neoantigen, no HLA LOH.
GSE65144	microarray	~13	ATC + normal	ATC enrichment	2	to_verify	Tiny n; suitable as ATC-vs-normal contrast in figure inset.
GSE54958	microarray	small	PTC stages	Stage-axis sensitivity	3	to_verify	
GSE53157	microarray	~27	PTC	Replication	3	to_verify	
GSE29265	microarray	~49	PTC, ATC, normal	Cross-platform	2	to_verify	
Yoo SK et al. 2019 *Nat Commun*	RNA-seq		Korean ATC + advanced DTC	ATC, advanced DTC			Asian dedifferentiation generalization

Full file: chmod 444 frozen, `project/reports/paper3_ici/paper3_ici_dataset_registry.md`

check supp: may be EGA-controlled. If controlled, add EGA application to wk 1.

2 Korean PTC | PTC | Asian baseline | 2 | to_verify | Cross-paper care – already touched by Paper 1/2 lineage. Use for HLA-typing reference, not for ecotype discovery, to keep Paper 3 distinct.

2. Signature Registry · `paper3_ici_signature_registry.md`

internal (read-only) | Per `v17\_arcashla\_korean\_k2`. Do not # Paper 3 ICI – Signature Registry
Paper 3 ICI – Signature Registry
Re-PUR; do not duplicate Paper 1/2 analyses. Use only as Korean HLA-frequency anchor. |
Working title: HLA loss, neoantigen architecture, and immune ecotypes define ICI vulnerability in molecularly dark thyroid cancer
Track: A – design only. No scoring runs. Gene members are listed for reference; final operational gene lists will be locked at Track B kickoff after intersection with each cohort's expression matrix.
Author: Seungho Cook
Status date: 2026-05-04

****Claim guard:**** Signatures are computed for ecotype discovery and DIAL audit, not as standalone "ICI response classifiers". Interpretation in thyroid is conditional on DIAL audit results (`paper3\_ici\_DIAL\_audit\_plan.md`).

0. Scoring conventions (locked at design)

- ****Bulk scoring method:**** ssGSEA (GSVA package) as default. Singscore as sensitivity (rank-based, sample-independent). Both reported in Supp.
- ****Single-cell scoring method:**** UCell (rank-based, robust to dropout) as default. AUCell as sensitivity.
- ****Normalization input:**** log2(TPM+1) for bulk RNA-seq; log-normalized counts (Seurat / scanpy default) for scRNA. Microarray cohorts use RMA-normalized log-intensity.
- ****Score sign convention:**** higher = more of the named feature. Direction-flips revealed by DIAL audit are documented per signature, never silently corrected.
- ****Cohort-level z-score:**** all signatures z-scored within cohort before cross-cohort comparison, to avoid platform-mean confounding. Pan-cohort scaling only applied after batch evaluation.
- ****Validation across platforms:**** any signature dropped if <50% gene members measurable on a microarray cohort; flagged in `paper3\_ici\_signature\_registry\_VERIFIED.tsv` at kickoff.

1. T-cell inflammation / IFN γ axis

Signature ID	Source	Members (representative)	Role	Notes
`IFNG_AYERS6`	Ayers et al. *JCI* 2017	IFNG, CXCL9, CXCL10, IDO1, HLA-DRA, STAT1	Core IFN γ axis	Used in pembrolizumab IFN γ -signature studies.
`IFNG_AYERS18`	(TIS, Tumor Inflammation Signature)	Ayers 2017 18 genes incl. CD8A, GZMK, CD27, LAG3, TIGIT, CXCL9, etc.	T-cell-inflamed phenotype	Dako commercial as nCounter; we re-implement as ssGSEA.
`CYTOLYTIC_ROONEY`	Rooney et al. *Cell* 2015	GZMA, PRF1	(geometric mean) Cytolytic activity	2-gene; very platform-stable.
`EFFECTOR_T`	Tirosh / Sade-Feldman composite	CD8A, CD8B, GZMB, GZMK, IFNG, NKG7, PRF1	Effector CD8	Bulk + scRNA.
`EXHAUSTED_T`	Tirosh / Wherry composite	PDCD1, CTLA4, LAG3, TIGIT, HAVCR2, TOX, TOX2, EOMES	Exhausted CD8	Co-occurs with IFNG axis; DIAL-flag candidate.

2. Antigen presentation machinery

Signature ID	Source	Members	Role	Notes
--- --- --- --- ---				
`MHC1_CORE`	composite	HLA-A, HLA-B, HLA-C, B2M, TAP1, TAP2, TAPBP, NLRC5, ERAP1	Class I machinery	Loss = immune escape.
`MHC2_CORE`	composite	HLA-DRA, HLA-DRB1, HLA-DPA1, HLA-DPB1, HLA-DQA1, HLA-DQB1, CIITA	Class II machinery	Tumor-intrinsic Class II expression in PTC is documented (HT-overlap signal in Paper 2). Important DIAL candidate.
`IFNG_RESPONSE_HALLMARK`	MSigDB Hallmark	hallmark IFNG_RESPONSE	Upstream context	Used as IFN γ -response sanity check.

📄 Full file: chmod 444 frozen, `project/reports/paper3_ici/paper3_ici_signature_registry.md`

3. scRNA Atlas Plan ·

`paper3_ici_scRNA_reference_atlas_summary.md`

```
# Paper 3 ICI – scRNA Reference Atlas Plan

**Working title:** HLA loss, neoantigen architecture, and
immune ecotypes define ICI vulnerability in molecularly dark
thyroid cancer
**Track:** A – design only. No integration runs. No Seurat /
scanpy / scVI / scANVI / Harmony invocation.
**Author:** Seungho Cook
**Status date:** 2026-05-04

**Claim guard:** Atlas defines cell states for *signature
derivation* and bulk projection. State frequencies in
patients are descriptive, not predictive of ICI response in
thyroid until DIAL audit + (where possible) thyroid ICI raw
RNA-seq supports such a claim.

---

## 1. Cohorts & expected scale

| Accession | Subtype skew | Expected cells (rough order) | Atlas role |
| ---|---|---|---|
| GSE184362 | PTC + ATC | 30-80K | Anchor – broadest cell-
type coverage |
| GSE193581 | PTC | 20-60K | Immune sub-states |
| GSE232237 | mixed | 20-50K | Integration validation |
| GSE191288 | PTC | 10-40K | Replication |
| GSE148673 | mixed | 10-30K | Replication |
```

| Han 2024 JCI Insight | dedifferentiation | 10-40K |
Dedifferentiation anchor |

****Total estimate:**** 100-300K cells after QC. Of these, PDTC + ATC fraction is expected to be small (<5K) – a critical caveat.

****Atlas-side risk register:****

- ****R1.**** PDTC/ATC scRNA cells <5K total → de novo dedifferentiation trajectory inference is underpowered. Mitigation: project pseudobulk-derived dedifferentiation signature from `paper3\_ici\_signature\_registry` Module 7 onto epithelial cells; do not run Slingshot/Monocle3 on ATC alone.
- ****R2.**** Sample-level batch effects dominate disease-level effects in thyroid scRNA. Mitigation: scVI with `batch\_key=sample\_id`, treat patient-level as donor-level; never use disease as integration covariate.
- ****R3.**** Thyrocyte epithelial cells often contaminated by adjacent normal in surgical specimens. Mitigation: copy-number-based malignant-cell calling (inferCNV / CopyKAT) before downstream epithelial-state analysis.

 Full file: chmod 444 frozen, `project/reports/paper3_ici/paper3_ici_scRNA_reference_atlas_summary.md`

4. HLA & Neoantigen Feasibility ·

`paper3_ici_HLA_neoantigen_feasibility.md`

Paper 3 ICI – HLA & Neoantigen Feasibility

****Working title:**** HLA loss, neoantigen architecture, and immune ecotypes define ICI vulnerability in molecularly dark thyroid cancer

****Track:**** A – design only. No HLA typing runs, no LOHHLA execution, no NetMHCpan invocation. arcasHLA results from `v17\_arcasHLA\_korean\_k2` are referenced read-only and ****not**** re-run.

****Author:**** Seungho Cook

****Status date:**** 2026-05-04

****Claim guard:**** HLA loss + neoantigen load are reported as **immunogenomic features**, not as standalone "ICI response signals". Combined into the integrated readiness score (`paper3\_ici\_signature\_registry.md` §9) only after DIAL audit.

1. Per-cohort feasibility matrix

```

| Cohort | RNA-seq | WES / WGS | HLA Class I | HLA Class II |
LOHHLA-feasible | Neoantigen-feasible | Notes |
|---|---|---|---|---|---|---|---|
| TCGA-THCA | yes | yes (BAM via GDC) | OptiType / Polysolver
from BAM | arcasHLA from RNA BAM | yes (paired tumor-normal
BAM) | yes (MAF + RNA expression filter) | Primary neoantigen
+ HLA LOH cohort. |
| GSE76039 PDTC/ATC (Landa 2016) | yes | yes (Landa 2016
published WES) | OptiType from RNA BAM (no germline normal) |
arcasHLA | partial – paired normal availability per-sample
needed | yes if MAF + paired normal | Verify normal
availability; without it, somatic calls less reliable. |
| Yoo SK 2019 Korean ATC | yes (RNA-seq) | likely yes
(matched WES) | arcasHLA | arcasHLA | conditional on access
mode | yes | EGA-controlled candidate; verify at Track B
kickoff. |
| PRJKA210106 Korean PTC n=282 | yes (RNA-seq) | unclear |
arcasHLA | arcasHLA | likely no (no paired normal) | no (no
MAF) | HLA frequency anchor only; cross-paper boundary care.
|
| PRJEB11591 Korean PTC n=260 | yes | no | arcasHLA already
done | arcasHLA already done | no | no | Read-only reference
per `v17_arcasHLA_korean_k2`. **Do not re-run.** |

**Effective Module C cohort = TCGA-THCA + GSE76039 + (Yoo
2019 if accessible).**

```

 Full file: chmod 444 frozen, `project/reports/paper3_ici/paper3_ici_HLA_neoantigen_feasibility.md`

5. DIAL Audit Plan · `paper3_ici_DIAL_audit_plan.md`

```

# Paper 3 ICI – DIAL Audit Plan

**Working title:** HLA loss, neoantigen architecture, and
immune ecotypes define ICI vulnerability in molecularly dark
thyroid cancer
**Track:** A – design only. No DIAL execution. No bootstrap,
no permutation, no Cox fitting.
**Author:** Seungho Cook
**Status date:** 2026-05-04

**Claim guard:** DIAL audit is the *gating step* that decides
which pan-cancer ICI signatures are allowed to enter Paper
3's integrated readiness score in thyroid. Without DIAL, the
integrated score has no validity argument.

---

## 1. Context – DIAL framework lineage

```

- DIAL = Direction-Invariance Algorithm. Originated in `v52\_lodo\_finding`: under proper LOD0 ComBat, all 5 THCA classifiers returned DIAL=0.000 / true biology, falsifying the v5.1 DIAL=0.494 flip artifact.
- Codified into `v18\_agentic\_research framework` (5 patterns + 6 components).
- Applied in Paper 3 to **pan-cancer ICI signatures** rather than to THCA classifiers – same algebra, different inputs.

Definition for Paper 3 use. A signature S is **direction-invariant** across context set C if its sign of association with the chosen endpoint (ICI response or T-cell-inflamed phenotype) is preserved across all leave-one-out subsets of C, with effect size > minimum. Failure modes:

- **Flip** – sign reverses in ≥ 1 leave-one-out fold.
- **Collapse** – effect size shrinks to noise ($|t| < 1$) in ≥ 1 fold.
- **Tissue-flip** – sign holds within trained tissue (e.g., melanoma) but reverses in held-out tissue (here: thyroid).

Only signatures that pass DIAL on the pan-cancer reference set are eligible to enter Paper 3's integrated score in thyroid.

 Full file: chmod 444 frozen, `project/reports/paper3_ici/paper3_ici_DIAL_audit_plan.md`

6. Figure Plan · `paper3_ici_figure_plan.md`

Paper 3 ICI – Figure Plan

Working title: HLA loss, neoantigen architecture, and immune ecotypes define ICI vulnerability in molecularly dark thyroid cancer

Track: A – design only. All figures are layout schematics. No real-data rendering. Captions are placeholder; final captions written during Wk 11.

Author: Seungho Cook

Status date: 2026-05-04

Claim guard: Each figure caption ends with a thyroid-ICI-vulnerability framing, never an "ICI response prediction" claim. Discrepancy between visualization and claim is a stop-the-press reviewer trigger; designer enforces this at draft.

1. Main figures (target = 6, ≤ 7)

```
### Figure 1 – Cohort landscape & dark-matter framing
- **1A** Schematic – PTC → PDTC → ATC dedifferentiation
continuum; BRAF / RAS / fusion / dark-matter color key;
immune-rich (HT-overlap, dark matter immune) overlay.
- **1B** Sample count matrix across cohorts × subtype (TCGA-
THCA, GSE76039, others) with modality availability (RNA /
WES / scRNA / spatial).
- **1C** TDS distribution by subtype + driver.
- **1D** Dark-matter (BRAF/RAS/fusion-negative) prevalence by
subtype.
- **1E** Cross-paper diagram showing Paper 1 / Paper 2 /
Paper 3 / Paper 4 scope boundaries – defensive against
reviewer scope-confusion.
```

```
### Figure 2 – Bulk immune ecotype discovery (Module A)
- **2A** Heatmap of standardized signature scores across
pooled bulk samples (rows = signatures from registry §1-§5,
cols = samples), NMF-derived ecotype labels.
- **2B** NMF rank selection (cophenetic correlation,
dispersion).
- **2C** Ecotype × subtype (PTC / PDTC / ATC) Sankey or
stacked bar.
- **2D** Ecotype × driver group (BRAF / RAS / fusion / dark
matter) stacked bar with chi-square.
- **2E** Ecotype × TDS tertile.
- **2F** Survival KM (PFS or DSS) by ecotype within
aggressive subset (TCGA-THCA stage III/IV + PDTC + ATC).
```

```
### Figure 3 – scRNA atlas & sub-state map (Module B)
### Figure 4 – HLA loss & neoantigen architecture (Module C)
### Figure 5 – DIAL audit (Module D)
### Figure 6 – Integrated thyroid ICI-readiness score (Module
E)
```

 Full file: chmod 444 frozen, `project/reports/paper3_ici/`
`paper3_ici_figure_plan.md`

7. Go / No-Go Verdict · `paper3_ici_go_no_go_verdict.md`

```
# Paper 3 ICI – Go / No-Go Verdict

**Working title:** HLA loss, neoantigen architecture, and
immune ecotypes define ICI vulnerability in molecularly dark
thyroid cancer
**Track:** A – design only. Verdict below is **conditional
GO** for Track B execution at the post-marathon entry point.
**Author:** Seungho Cook
**Status date:** 2026-05-04

**Claim guard:** Below verdict assumes Paper 3 is framed as
ICI vulnerability / readiness / immunogenomic prioritization.
```

If the user later wants to claim "ICI response prediction" without thyroid ICI-treated raw RNA-seq, verdict drops to NO-GO and must be re-decided.

1. Module-level verdicts

Module	Verdict	Rationale	Critical dependency
A – Bulk immune ecotype	**GO**	TCGA-THCA + GSE76039 + microarray cohorts give ≥600 samples after pooling; ssGSEA + NMF mature; thyroid ecotypes literature-supported.	None blocking.
B – scRNA atlas	**CONDITIONAL GO**	6 candidate cohorts, total ~100-300K cells. PDTC/ATC scRNA <5K – atlas viable for state annotation, not for de novo dedifferentiation trajectory. Mitigation framed in atlas plan §1. At least 3 of 6 cohorts must be accessible at Wk 5.	
C – HLA & neoantigen	**GO (degraded scope)**	TCGA-THCA paired BAM + GDC MC3 MAF available; LOHHLA + NetMHCpan well-established; GSE76039 paired-normal access uncertain per sample; Yoo 2019 may be EGA-controlled.	TCGA-THCA dbGaP controlled-access for paired BAM by Wk 1 of Track B.
D – DIAL audit	**GO**	Pan-cancer ICI cohorts mostly accessible (IMvigor210 R-package; Hugo / Riaz / Gide GEO; Liu / Cho controlled). Statistical machinery already in `v18_agentic_research` framework.	At least 3 of 5 Tier 1/2 melanoma + UC cohorts accessible by Wk 7.
E – Integrated readiness score	**GO conditional on D**	Score is well-defined; weighting calibration deferred.	DIAL audit must produce ≥4 PASS signatures for the score to be meaningful.

Full file: `chmod 444 frozen, project/reports/paper3_ici/paper3_ici_go_no_go_verdict.md`
Overall verdict – Conditional GO. Paper 3 is feasible. Pass events that map to the 12-week plan's Wk 1 / Wk 5 / Wk 7 gates, not analytical uncertainties.

8. 12-Week Execution Plan ·

`paper3_ici_12week_execution_plan.md`

Paper 3 ICI – 12-Week Execution Plan

Working title: HLA loss, neoantigen architecture, and immune ecotypes define ICI vulnerability in molecularly dark thyroid cancer

Track: A (this document) – design only. Plan below describes Track B execution; Track B starts only after entry gates G1-G6 close.

Author: Seungho Cook

Status date: 2026-05-04

****Claim guard:**** All weeks below assume "ICI vulnerability / readiness / immunogenomic prioritization" framing. If thyroid ICI-treated raw RNA-seq is acquired during the plan, framing may upgrade – only after re-decision with user, never silently.

0. Entry preconditions (G1-G6)

Plan does not start until:

- G1 Paper 1 bioRxiv submitted (target 2026-06-13)
- G2 Paper 2 Task A/B/C closed
- G3 TCGA dbGaP access decided
- G4 ≥ 3 of 6 scRNA cohorts accessible
- G5 ≥ 3 of 5 Tier 1/2 pan-cancer ICI cohorts accessible
- G6 User explicit "Track B 시작"

If G1/G2 not closed, the plan re-baselines its Wk 1 day to the post-ship date.

1. Week-by-week schedule

Wk 1 – Dataset acquisition & ETL scaffolding

- Verify all `to\_verify` accessions in `paper3\_ici\_dataset\_registry.md` §1-§4. Produce `paper3\_ici\_dataset\_registry\_VERIFIED.tsv`.
- Submit dbGaP / EGA applications in parallel (Liu melanoma, Yoo 2019 if EGA, TCGA paired-BAM controlled if needed).
- Stand up `project/results/paper3\_ici/` directory tree per registry / atlas / hla / neo / dial / integrated outputs.

 Full file: `chmod 444 frozen, project/reports/paper3_ici/paper3_ici_12week_execution_plan.md` Fully-measurable subset per cohort. Produce `paper3\_ici\_signature\_registry\_VERIFIED.tsv`.

Wk 2 – Bulk preprocessing & cohort harmonization

Wk 3 – Bulk immune ecotype discovery (Module A part 1)

Wk 4 – Bulk ecotype × clinical/molecular axes (Module A part 2)

Wk 5 – scRNA atlas integration (Module B part 1)

Wk 6-10 – Module C HLA + neoantigen, Module D DIAL audit, Module E integrated readiness score

Wk 11-12 – Manuscript draft

13. 한국어 설명 (방법 · 결과 · 의미 풀어쓰기)

13.1 왜 thyroid cancer 에 ICI vulnerability 가 의미 있 나

전체 PTC 의 ~95% 는 surgical resection + RAI 치료로 cure 되지만 ~4% 는 high-risk recurrent / metastatic 으로 진행. 이 4% 안에 (a) **BRAF V600E + TERT** driver-positive aggressive type, (b) **BRAF/RAS-negative dark matter** driver-unknown type 이 같이 있다. Driver-positive 는 BRAF inhibitor (dabrafenib + trametinib) 등 targeted therapy 가 후보. **Driver-negative dark matter** 환자는 targeted therapy 후보 없음 – 면역치료 (PD-1 / PD-L1 inhibitor) 가 마지막 옵션. 그러나 thyroid 는 (a) tumor mutational burden 평균 낮음, (b) MHC-I / MHC-II 발현 cohort heterogeneity, (c) ICI clinical trial 매우 적음 – ICI 후보 집단 정의 자체가 unsolved.

13.2 왜 "ICI vulnerability" 만 OK 이고 "ICI response predictor" 는 K1 kill 인가

"Response predictor" 는 ICI 받은 환자의 outcome 을 예측하는 model – 이걸 만들려면 **ICI-treated thyroid cancer cohort raw RNA-seq + treatment outcome** 데이터가 필요. 전세계 부재. 따라서 어떤 signature 가 thyroid 의 ICI response 를 예측한다고 못 한다. 대신 "**vulnerability profile**" = HLA loss / neoantigen burden / immune ecotype 의 분자 marker 만으로 ICI 후보 prioritize. 이걸 published immune-oncology 의 framework 안에서 (TIDE Jiang 2018 / IMPRES Auslander 2018 / TIS Ayers 2017 / etc.) thyroid-specific 적용. **K1 kill switch** 는 manuscript / cover letter / abstract 어디에도 "predictor" 라는 단어 사용 시 즉시 paper 파기.

13.3 왜 chmod 444 frozen 인가

Track A (디자인 phase) 는 12 주 execution 의 청사진 - 어떤 cohort 어떤 signature 어떤 method 어떤 figure 어떤 verdict matrix. 이 디자인 자체가 manuscript 의 backbone – 만약 marathon 중 design 을 마음대로 수정하면 (a) reviewer Q1 "왜 ICI predictor 라고 안 했나" 의 사전 답변이 흔들림, (b) signature DIAL audit plan 이 post-hoc selection bias 가능, (c) scope creep 으로 Paper 1 마라톤 displacement. **chmod 444 + tar.gz + sha256** 으로 lock – Track B unlock 시까지 read-only. unlock 조건 3개 모두 충족 (Paper 1 bioRxiv + Paper 2B Pillar II/III + 명시적 명령) 시에만 chmod 644 로 변경.

I4. 이 연구의 한계 (구체적, 솔직하게)

1. **thyroid ICI raw cohort 부재 (전세계).** 이건 paper 의 K1 kill switch 의 근거 – "predictor" 라고 못 하는 핵심 이유. 일부 thyroid PD-1 trial (NCT04645966 등) 은 결과 미공개 / cohort 작음 / 전공조회 controlled access 필요.
2. **Pan-cancer ICI signatures 의 thyroid sign-flip 가능성.** TIDE Jiang 2018 / IMPRES Auslander 2018 / TIS Ayers 2017 등은 melanoma / NSCLC / urothelial 에서 검증 – thyroid 에서 sign 이 같은 방향인지 미검증. K5 kill switch 가 이 risk 를 명시적으로 보호.
3. **dbGaP / EGA controlled access 신청 + 응답 시간.** Pan-cancer ICI cohort (Liu 2019 / Riaz 2017 / Hugo 2016) 대부분 controlled access – 신청 후 응답 3-6개월. 12-week execution 안에 못 맞출 가능성.
4. **scRNA atlas 의 PDTC/ATC undersampled.** Pu 2021 (PMC8754658) 도 PDTC/ATC < 5K cell – sc-derived signature 가 underpowered. K2 kill switch.
5. **cookHLA 사용 못함 (Paper 4 reserved).** Paper 3 의 HLA module 은 OptiType / Polysolver / arcasHLA / xHLA 4가지 imputation consensus 만 – cookHLA 의 cross-disease 강점 활용 못 함. 이건 Paper 4 boundary discipline.
6. **integrated readiness score 의 reproducibility.** G6 hard gate (cross-cohort $r \geq 0.6$) 가 통과해야 publish 가능. Pan-cancer ICI cohort 의 thyroid transfer 가 약하면 score 정의 자체가 흔들림.
7. **"vulnerability" 가 임상적으로 actionable 하지 않을 수 있음.** ICI clinical trial 등록 기준 / treatment outcome 측정 / safety profile 등 임상 transition 은 이 paper 의 scope 외.

14.I 다음 행동 (post-Track B unlock)

- Manuscript draft → Yu review → Cell Rep Med / JCI Insight 제출
~2026-12 target (post-Paper 1 + 2B)
- cookHLA 사용 unblock 시 (Paper 4 이후) → HLA module 강화 가능

15. Claim boundary table

CLAIM	STATUS
Track A 8 deliverables chmod 444 frozen 2026-05-04	YES
tar.gz 51 KB · sha256 290349f6...8936aa archive	YES
"ICI vulnerability" / "ICI-readiness" / "immunogenomic prioritization" framing	ALLOWED
"ICI response predictor" claim (K1)	FORBIDDEN – kill
v15 v5.1 DIAL = 0.494 number cite (K2 post-v52 lock)	FORBIDDEN – kill
Paper 1/2B scope contamination (K3)	FORBIDDEN – kill
Track B execution during marathon	BLOCKED
Track B 12-week manuscript draft	FUTURE (3-condition unlock)
cookHLA usage in Paper 3	RESERVED for Paper 4

16. 데이터 / 다운로드

16.1 Anchor files

ROLE	PATH
Freeze notice	<code>project/reports/ PAPER3_ICI_TRACK_A_FREEZE_2026_05_04.md</code>
Decision brief	<code>project/reports/ PAPER3_ICI_DECISION_BRIEF_2026_05_04.md</code> (A/B/C tree)
Single-file bundle (chmod 444)	<code>project/reports/paper3_ici/ PAPER3_ICI_TRACK_A_BUNDLE.md</code> · 1236 lines
Tar.gz archive	<code>project/reports/paper3_ici/ paper3_ici_track_a.tar.gz</code> · 51 KB · sha256 290349f6...8936aa
8 individual designs	<code>project/reports/paper3_ici/paper3_ici_*.md</code> (chmod 444)
Editable todo	<code>project/reports/paper3_ici/todo/</code> (chmod 644 · 3 checklists)

16.2 Cross-paper bridges

Paper 1 = molecular dark matter envelope (DM1/DM2). Paper 2B = HT-overlap 면역 microenvironment characterization. Paper 3 = ICI-readiness layer (Paper 1 + Paper 2B 위에). Paper 4 = cookHLA Korean GD HLA (Paper 3의 HLA module 분리). Paper 7 Component B (chmod 444 audit-bundle freeze) example.

16.3 Quick links

← PAPER 1

Molecular dark matter
envelope

← PAPER 2B

HT-overlap PTC
immune
characterization

→ PAPER 4

Korean GD HLA /
Pan-Asian (backlog)

→ PAPER 6

v52_lodo_finding
lesson home

→ PAPER 7

Component B chmod
444 archetype

 HUB INDEX

8-papers portfolio

Status. Paper 3 = ICI vulnerability dark thyroid cancer · Track A FROZEN
2026-05-04 (chmod 444 · 8 deliverables · 1,236 lines · tar.gz 51 KB sha256
290349f6...8936aa) · Track B BLOCKED until Paper 1 bioRxiv + Paper 2B Pillar II/III
closure + literal "Paper 3 Track B 시작" command. Per memory
v19_paper3_ici_track_a .

Live URL. http://40.82.129.113:8012/papers_hub_2026_05_04/paper3.html